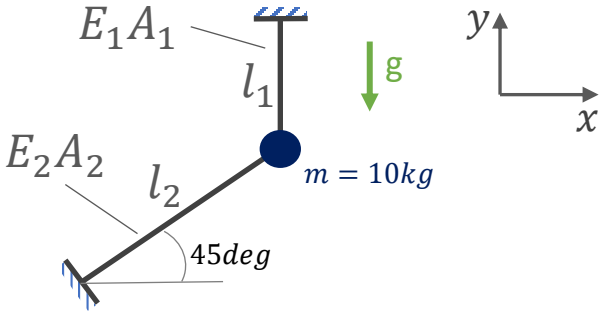
Structures and Materials

Tutorial | Dr.-Ing. Andreas Hermanutz

Finite Element Method

1. FEM - Framework



Task 1: Consider the truss framework shown on the right. Determine the vertical deflection of the mass m due to gravity ($g = 9.81\ \mathrm{m/s^2}$).

The framework consists of two truss members:

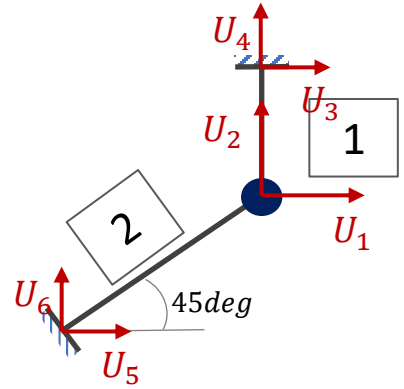
- **Truss 1:** Titanium with a Young's modulus of $E = 110\ \mathrm{Gpa}$, $l_1 = 200mm$.
- **Truss 2:** Quasi-isotropic GFRP with a Young's modulus of $E = 24\ \mathrm{GPa}$, $l_4 = 400mm$.

Both truss members have a uniform cross-sectional area of $A = 100\ \mathrm{mm^2}$.

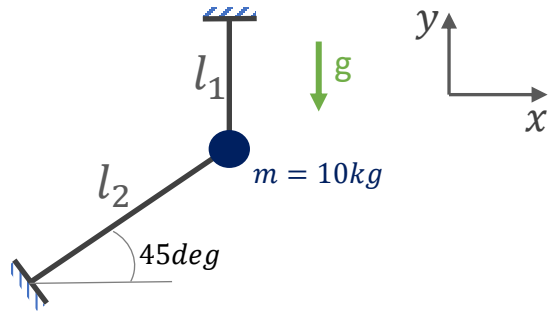
Step 1: Calculate the stiffness matrix of the rod l_1 and l_2

$$k_1 = \frac{EA}{l_1}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}\begin{matrix} U_2 \\ U_4 \end{matrix} \qquad \begin{matrix} U_2 & U_4 \end{matrix}$$

$$k_2 = \frac{EA}{l_2}\begin{bmatrix} c^2 & sc & -c^2 & -sc \\ sc & s^2 & -sc & -s^2 \\ -c^2 & -sc & c^2 & sc \\ -sc & -s^2 & sc & s^2 \end{bmatrix} = \frac{EA}{l_2}\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \frac{EA}{2l_2}\begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{bmatrix}\begin{matrix} U_5 \\ U_6 \\ U_1 \\ U_2 \end{matrix} \qquad \begin{matrix} U_5 & U_6 & U_1 & U_2 \end{matrix}$$



1. FEM - Framework



Task 1: Consider the truss framework shown on the right. Determine the vertical deflection of the mass m due to gravity ($g = 9.81\ \mathrm{m/s^2}$).

Step 2: Calculate the global stiffness matrix.

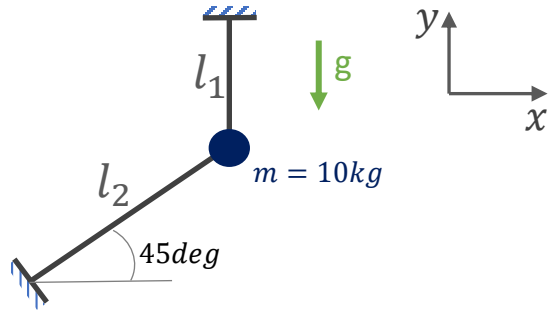
$$k_1 = \frac{EA}{l_1}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{matrix} U_2 \\ U_4 \end{matrix} \qquad \begin{matrix} U_2 & U_4 \end{matrix}$$

$$k_2 = \frac{EA}{2l_2}\begin{bmatrix} 1 & 1 & -1 & -1 \\ 1 & 1 & -1 & -1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & 1 & 1 \end{bmatrix} \begin{matrix} U_5 \\ U_6 \\ U_1 \\ U_2 \end{matrix} \qquad \begin{matrix} U_5 & U_6 & U_1 & U_2 \end{matrix}$$

$$K = \begin{bmatrix} \dfrac{E_2A_2}{2l_2} & \dfrac{E_2A_2}{2l_2} \\ \dfrac{E_2A_2}{2l_2} & \dfrac{E_1A_1}{l_1} + \dfrac{E_2A_2}{2l_2} \end{bmatrix} \begin{matrix} U_1 \\ U_2 \end{matrix} \qquad \begin{matrix} U_1 & U_2 \end{matrix}$$

1. FEM - Framework

Task 1: Consider the truss framework shown on the right. Determine the vertical deflection of the mass m due to gravity ($g = 9.81\ \mathrm{m/s^2}$).



Step 3: Calculate the applied load vector.

$$f = \begin{bmatrix} 0 \\ -m \cdot g \end{bmatrix} = \begin{bmatrix} 0 \\ \mathbf{-98.1} \end{bmatrix}$$

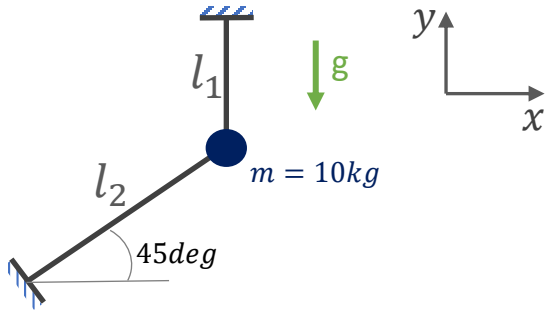
$$\boldsymbol{K} = \begin{bmatrix} \mathbf{3000} & \mathbf{3000} \\ \mathbf{3000} & \mathbf{58000} \end{bmatrix} \quad ! \ \mathrm{N/mm}$$

Step 4: Solve the system $[K]\{u\} = \{f\}$.

$$\{u\} = \frac{1000}{58000 \cdot 3000 - 3000^2} \begin{bmatrix} 58 & -3 \\ -3 & 3 \end{bmatrix} \begin{Bmatrix} \mathbf{0} \\ \mathbf{-98.1} \end{Bmatrix} = \begin{Bmatrix} \mathbf{0.001784} \\ \mathbf{-0.001784} \end{Bmatrix} \boldsymbol{mm}$$

1. FEM - Framework

Task 2: Consider the truss framework shown on the right. Calculate the reaction forces.



$$\left[\begin{array}{c|c} K_{11} & K_{12} \\ \hline K_{21} & K_{22} \end{array}\right] \left\{\begin{array}{c} \boldsymbol{U_1} \\ \boldsymbol{U_2} \\ \hline \boldsymbol{U_3 = 0} \\ \boldsymbol{U_4 = 0} \\ \boldsymbol{U_5 = 0} \\ \boldsymbol{U_6 = 0} \end{array}\right\} = \begin{Bmatrix} \boldsymbol{f_1} \\ \boldsymbol{f_2} \\ \boldsymbol{F_3} \\ \boldsymbol{F_4} \\ \boldsymbol{F_5} \\ \boldsymbol{F_6} \end{Bmatrix}$$

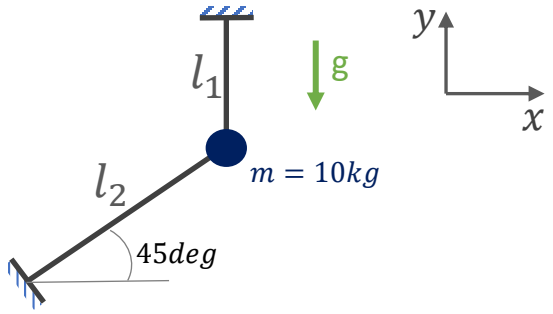
$$K_{12} = \begin{array}{c} \begin{matrix} U_3 & U_4 & U_5 & U_6 \end{matrix} \\ \begin{bmatrix} 0 & 0 & -\frac{EA}{l_2} & -\frac{EA}{l_2} \\ 0 & -\frac{EA}{l_1} & -\frac{EA}{l_2} & -\frac{EA}{l_2} \end{bmatrix} \end{array} \begin{matrix} U_1 \\ U_2 \end{matrix} = \begin{bmatrix} 0 & 0 & -3000 & -3000 \\ 0 & -55000 & -3000 & -3000 \end{bmatrix}$$

$$[K_{21}] \begin{Bmatrix} \boldsymbol{U_1} \\ \boldsymbol{U_2} \end{Bmatrix} + [K_{22}] \begin{Bmatrix} \boldsymbol{U_3} \\ \boldsymbol{U_4} \\ \boldsymbol{U_5} \\ \boldsymbol{U_6} \end{Bmatrix} = \begin{Bmatrix} \boldsymbol{F_3} \\ \boldsymbol{F_4} \\ \boldsymbol{F_5} \\ \boldsymbol{F_6} \end{Bmatrix}$$

$$\begin{Bmatrix} \boldsymbol{F_3} \\ \boldsymbol{F_4} \\ \boldsymbol{F_5} \\ \boldsymbol{F_6} \end{Bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & -55000 \\ -3000 & -3000 \\ -3000 & -3000 \end{bmatrix} \begin{Bmatrix} \boldsymbol{0.0017836} \\ \boldsymbol{-0.0017836} \end{Bmatrix} = \begin{Bmatrix} \boldsymbol{0} \\ \boldsymbol{98.1} \\ \boldsymbol{0} \\ \boldsymbol{0} \end{Bmatrix} \boldsymbol{N}$$

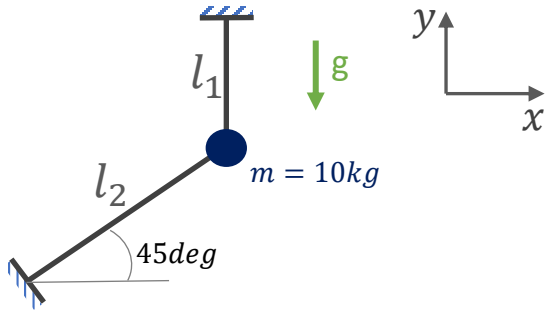
1. FEM - Framework

Task 3: Calculate the Stress in the truss.



$$\sigma = E \cdot \varepsilon = \frac{E}{l}[-1 \quad 1]\begin{Bmatrix} d_1 \\ d_2 \end{Bmatrix} = \frac{110000}{200}[-1 \quad 1]\begin{Bmatrix} \mathbf{-0.0017836} \\ 0 \end{Bmatrix} = 0.98098\ MPa$$

1. FEM - Framework



Task 4: The displacement of the mass is specified to be **1 mm in the x-direction**. What additional load, besides the gravitational load, must be applied to the mass (m) to achieve this displacement?

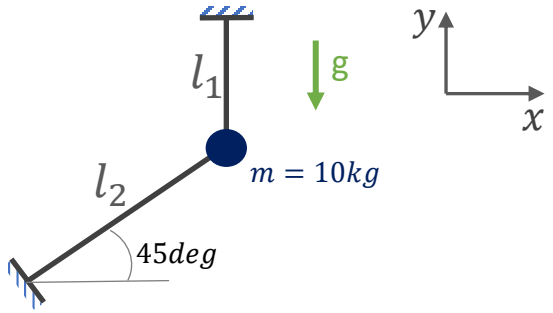
$$\begin{bmatrix} \mathbf{3000} & \mathbf{3000} \\ \mathbf{3000} & \mathbf{58000} \end{bmatrix} \begin{Bmatrix} U_1 = 1 \\ U_2 \end{Bmatrix} = \begin{Bmatrix} \boldsymbol{F} \\ \mathbf{-98.1} \end{Bmatrix}$$

$$U_2 = \frac{1}{58000}(-98.1 - 3000) = -0.0534155\, mm$$

$$F = 3000 + 3000 \cdot U_2 = 3000 - 3000 \cdot 0.0534155 = 2.83975kN$$

1. FEM - Framework

Task 5: Calculate the eigenfrequencies of the structure using a lumped mass matrix, assuming that the two truss elements are massless.



$$\boldsymbol{K} = \begin{bmatrix} \mathbf{3000} & \mathbf{3000} \\ \mathbf{3000} & \mathbf{58000} \end{bmatrix} \qquad \boldsymbol{M} = \begin{bmatrix} \mathbf{10} & \mathbf{0} \\ \mathbf{0} & \mathbf{10} \end{bmatrix}$$

$$det([\boldsymbol{k}] - \lambda[\boldsymbol{m}]) = 0 \qquad \lambda = \omega^2$$

$$det\left(\begin{bmatrix} \mathbf{3000} - \lambda \cdot 10 & \mathbf{3000} \\ \mathbf{3000} & \mathbf{58000} - \lambda \cdot 10 \end{bmatrix}\right) = 0$$

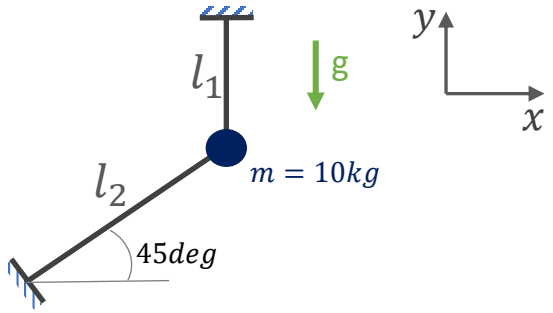
$$(\mathbf{3000} - \lambda \cdot 10) \cdot (\mathbf{58000} - \lambda \cdot 10) - \mathbf{3000^2} = \mathbf{0}$$

$$\lambda_1 = 283.685 \qquad \lambda_2 = 5816.32$$

$$\omega_1 = \sqrt{\lambda_1} = 532.62\ rad/s \qquad \omega_2 = \sqrt{\lambda_2} = 2411.7\ rad/s$$

$$f_1 = \frac{\omega_1}{2\pi} = 84.7\ Hz \qquad f_2 = \frac{\omega_2}{2\pi} = 383.8\ Hz$$

1. FEM - Framework



Task 6: Calculate the eigenvector and create a sketch of the eigenform.

$$\boldsymbol{K} = \begin{bmatrix} \mathbf{3000} & \mathbf{3000} \\ \mathbf{3000} & \mathbf{58000} \end{bmatrix}$$

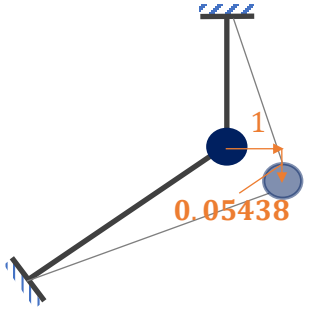
$$\boldsymbol{M} = \begin{bmatrix} \mathbf{10} & \mathbf{0} \\ \mathbf{0} & \mathbf{10} \end{bmatrix}$$

$$det([\boldsymbol{k}] - \lambda[\boldsymbol{m}]) = 0$$

$\lambda_1 = 2.83685$:

$$\begin{bmatrix} \mathbf{3000} - \lambda_1 \cdot 10 & \mathbf{3000} \\ \mathbf{3000} & \mathbf{58000} - \lambda_1 \cdot 10 \end{bmatrix} = \begin{bmatrix} \mathbf{163.15} & \mathbf{3000} \\ \mathbf{3000} & \mathbf{55163.15} \end{bmatrix}$$

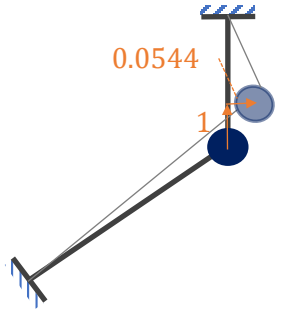
$$\begin{bmatrix} \mathbf{163.15} & \mathbf{3000} \\ \mathbf{3000} & \mathbf{55163.15} \end{bmatrix} \begin{Bmatrix} \boldsymbol{\phi_{11}} \\ \boldsymbol{\phi_{12}} \end{Bmatrix} = \mathbf{0} \qquad \begin{Bmatrix} \boldsymbol{\phi_{11}} \\ \boldsymbol{\phi_{12}} \end{Bmatrix} = \begin{Bmatrix} \mathbf{1} \\ \mathbf{-0.05438} \end{Bmatrix}$$



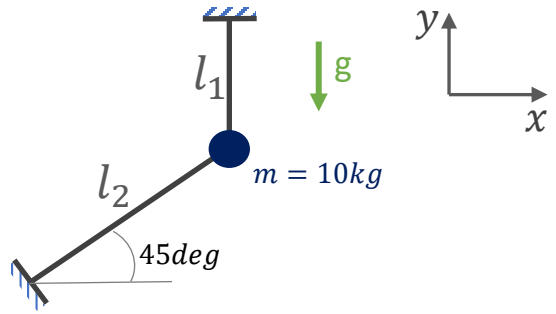
$\lambda_2 = 5.8132$

$$\begin{bmatrix} \mathbf{3000} - \lambda_2 \cdot 10 & \mathbf{3000} \\ \mathbf{3000} & \mathbf{58000} - \lambda_2 \cdot 10 \end{bmatrix} = \begin{bmatrix} \mathbf{-55163.15} & \mathbf{3000} \\ \mathbf{3000} & \mathbf{-163.15} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{-55163.15} & \mathbf{3000} \\ \mathbf{3000} & \mathbf{-163.15} \end{bmatrix} \begin{Bmatrix} \boldsymbol{\phi_{21}} \\ \boldsymbol{\phi_{22}} \end{Bmatrix} = \mathbf{0} \qquad \begin{Bmatrix} \boldsymbol{\phi_{21}} \\ \boldsymbol{\phi_{22}} \end{Bmatrix} = \begin{Bmatrix} \mathbf{0.05438} \\ \mathbf{1} \end{Bmatrix}$$



1. FEM - Framework



Task 7: Transform the system into generalized coordinates while accounting for gravitational loading.

$$\boldsymbol{\phi} = \begin{bmatrix} 1 & 0.05438 \\ -0.05438 & 1 \end{bmatrix}$$

$$\boldsymbol{K_{gen}} = \boldsymbol{\phi}^T[K]\boldsymbol{\phi} = \begin{bmatrix} 1 & -0.05438 \\ 0.05438 & 1 \end{bmatrix}\begin{bmatrix} 3000 & 3000 \\ 3000 & 58000 \end{bmatrix}\begin{bmatrix} 1 & 0.05438 \\ -0.05438 & 1 \end{bmatrix} = \begin{bmatrix} 2845.24 & 0.0 \\ 0 & 58335.18 \end{bmatrix} \text{ N/mm}$$

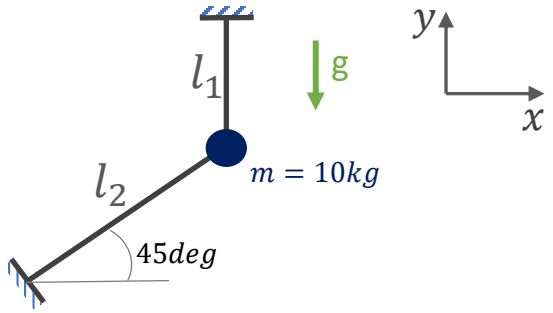
$$\boldsymbol{M_{gen}} = \boldsymbol{\phi}^T[M]\boldsymbol{\phi} = \begin{bmatrix} 1 & -0.05438 \\ 0.05438 & 1 \end{bmatrix}\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}\begin{bmatrix} 1 & 0.05438 \\ -0.05438 & 1 \end{bmatrix} = \begin{bmatrix} 0.0100296 & 0.0 \\ 0 & 0.0100296 \end{bmatrix}$$

$$\boldsymbol{F_{gen}} = \boldsymbol{\phi}^T \begin{Bmatrix} 0 \\ -98.1 \end{Bmatrix} = \begin{Bmatrix} 5.335 \\ -98.1 \end{Bmatrix}$$

$$\boldsymbol{q} = \begin{Bmatrix} q_1 \\ q_2 \end{Bmatrix}$$

$$\begin{bmatrix} 0.0100296 & 0.0 \\ 0 & 0.0100296 \end{bmatrix}\ddot{\boldsymbol{q}} + \begin{bmatrix} 2845.24 & 0.0 \\ 0 & 58335.18 \end{bmatrix}\boldsymbol{q} = \begin{Bmatrix} 5.335 \\ -98.1 \end{Bmatrix} \qquad \boldsymbol{q} = \begin{Bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{Bmatrix}$$

1. FEM - Framework



Task 8: Perform a mass normalization of the mass matrix.

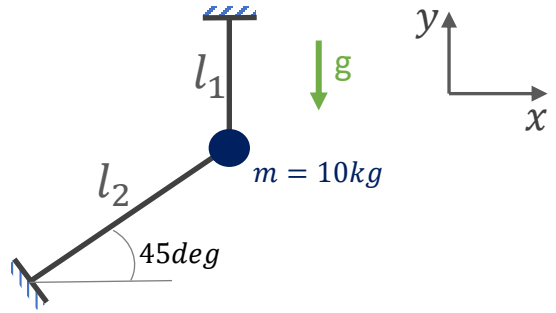
$$\boldsymbol{\phi} = \begin{bmatrix} 1 & 0.05438 \\ -0.05438 & 1 \end{bmatrix}$$

$$\tilde{M} = \boldsymbol{\phi}_1^T \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} \boldsymbol{\phi}_1 = \begin{bmatrix} 1 & -0.05438 \end{bmatrix} \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ -0.05438 \end{bmatrix} = 10.029576 \qquad \alpha_1 = \sqrt{10.029576} = 3.16696$$

$$\tilde{M} = \boldsymbol{\phi}_2^T \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} \boldsymbol{\phi}_2 = \begin{bmatrix} -0.05438 & 1 \end{bmatrix} \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} -0.05438 \\ 1 \end{bmatrix} = 10.029576 \qquad \alpha_2 = \sqrt{10.029576} = 3.16696$$

$$\hat{\boldsymbol{\phi}}_1 = \frac{1}{3.16696} \begin{bmatrix} 1 \\ -0.05438 \end{bmatrix} \qquad \hat{\boldsymbol{\phi}}_2 = \frac{1}{3.16696} \begin{bmatrix} 0.05438 \\ 1 \end{bmatrix} \qquad \hat{\boldsymbol{\phi}} = \begin{bmatrix} 0.315761 & 0.017172 \\ -0.017172 & 0.315761 \end{bmatrix}$$

1. FEM - Framework



Task 9: Transform the system into generalized coordinates while accounting for gravitational loading using a mass normalized matrix.

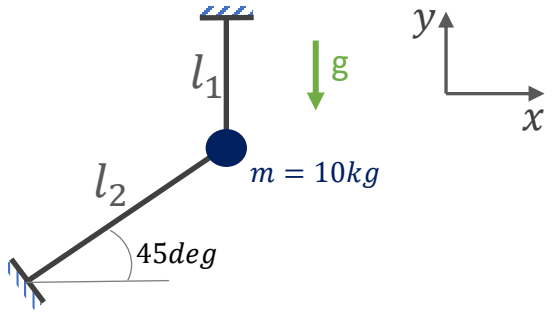
$$K_{gen} = \phi^T[K]\phi = \begin{bmatrix} 0.315761 & -0.017172 \\ 0.017172 & 0.315761 \end{bmatrix}\begin{bmatrix} 3000 & 3000 \\ 3000 & 58000 \end{bmatrix}\begin{bmatrix} 0.315761 & 0.017172 \\ -0.017172 & 0.315761 \end{bmatrix} = \begin{bmatrix} 283.685 & 0.0 \\ 0 & 5816.32 \end{bmatrix} \text{ N/mm}$$

$$M_{gen} = \phi^T[M]\phi = \begin{bmatrix} 0.315761 & -0.017172 \\ 0.017172 & 0.315761 \end{bmatrix}\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}\begin{bmatrix} 0.315761 & 0.017172 \\ -0.017172 & 0.315761 \end{bmatrix} = \begin{bmatrix} 1.0 & 0.0 \\ 0 & 1.0 \end{bmatrix}$$

$$F_{gen} = \begin{bmatrix} 0.315761 & -0.017172 \\ 0.017172 & 0.315761 \end{bmatrix}\begin{Bmatrix} 0 \\ -98.1 \end{Bmatrix} = \begin{Bmatrix} -1.685 \\ -30.976 \end{Bmatrix}$$

$$\begin{bmatrix} 1.0 & 0.0 \\ 0 & 1.0 \end{bmatrix}\ddot{q} + \begin{bmatrix} \lambda_1 & 0.0 \\ 0 & \lambda_2 \end{bmatrix}q = \begin{Bmatrix} -1.685 \\ -30.976 \end{Bmatrix}$$

1. FEM - Framework



Task 10: Calculate the mass matrix of the system using a consistent mass formulation. Use a density of $\rho_{Ti} = 4430\ \mathrm{kg}/m^3$ for titanium and $\rho_{GFRP} = 1800\ \mathrm{kg}/m^3$ for glass-fibre-reinforced polymer (GFRP).

$$M_1 = \frac{\rho A l}{6}\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix} = \begin{matrix}\\ U_2\\ U_4\end{matrix}\begin{matrix}\begin{matrix}U_2 & U_4\end{matrix}\\ \begin{bmatrix}0.029533 & 0.014767\\ 0.014767 & 0.029533\end{bmatrix}\end{matrix}$$

$$M_2 = \rho A l\begin{bmatrix}\frac{1}{3}c^2 & \frac{1}{3}sc & \frac{1}{6}c^2 & \frac{1}{6}sc\\ \frac{1}{3}sc & \frac{1}{3}s^2 & \frac{1}{6}sc & \frac{1}{6}s^2\\ \frac{1}{6}c^2 & \frac{1}{6}sc & \frac{1}{3}c^2 & \frac{1}{3}sc\\ \frac{1}{6}sc & \frac{1}{6}s^2 & \frac{1}{3}sc & \frac{1}{3}s^2\end{bmatrix} = \begin{matrix}\\ U_5\\ U_6\\ U_1\\ U_2\end{matrix}\begin{matrix}\begin{matrix}U_5 & U_6 & U_1 & U_2\end{matrix}\\ \begin{bmatrix}0.012 & 0.012 & 0.006 & 0.006\\ 0.012 & 0.012 & 0.006 & 0.006\\ 0.006 & 0.006 & 0.012 & 0.012\\ 0.006 & 0.006 & 0.012 & 0.012\end{bmatrix}\end{matrix}$$

$$\boldsymbol{M} = \begin{bmatrix}\mathbf{0.012 + 10} & \mathbf{0.012}\\ \mathbf{0.012} & \mathbf{0.029533 + 0.012 + 10}\end{bmatrix}$$

$$\boldsymbol{M} = \begin{bmatrix}\mathbf{10.012} & \mathbf{0.012}\\ \mathbf{0.012} & \mathbf{10.041533}\end{bmatrix}$$